

NOLAN KNIGHT

Evanston, IL | nolanknight133@icloud.com | (574) 453-7472
github.com/ncknight-un | linkedin.com/in/nolan-knight-nu | nolan-knight.com

EDUCATION

Northwestern University - Evanston, IL Master of Science in Robotics	Expected December 2026
Purdue University - Weldon School of Biomedical Engineering - West Lafayette, IN Master of Science in Biomedical Engineering	May 2025
Butler University - Indianapolis, IN Bachelor of Arts in Computer Science	May 2024
Purdue School of Engineering & Technology - Indianapolis, IN Bachelor of Science in Biomedical Engineering	May 2024

WORK EXPERIENCE

Southwest Research Institute , Graduate Student Robotics Intern	June 2026 - Current
- Developed a PyBullet simulation integrating physics and feedback control to validate robotic system behavior. - Integrated a joystick controller to enable real-time, hands-on control of a simulated robot in a virtual environment. - Designed a preliminary docking station in SolidWorks for an ongoing robotics project. - Supported testing and validation workflows across multiple robotics projects, including ROS 2 integration.	
GM Diecron Inc. , CAD/Automation Manufacturing (Contract) - Remote	May 2025 - August 2025
- Designed and modeled robotic machining layouts using Fusion 360 for aerospace manufacturing applications. - Modeled and assembled 3D designs and milling fixtures per engineering drawings and design constraints. - Produced fixture assemblies, orientation zones, and motion studies for robotic workflow automation.	
Zimmer Biomet , Development Engineering Co-Op - Warsaw, IN	May 2024 - August 2024
- Developed verification protocols and refined ROM reports to ensure compliance with ASTM / ISO Standards. - Improved anatomical knowledge through initiating involvement in sawbones, specimen harvesting, and cadaver labs.	

PROJECTS

Custom Vision-ML & Control Stack for BlueROV - Sealbot	Spring 2026 - Current
- Developing an onboard ROS 2 control stack for detection, image processing, and feedback control with BlueROV. - Developed a vision-ML system using a Jetson Orin Nano and Alvium camera for onboard image processing. - Designed and machined a waterproof aluminum enclosure to protect and integrate the embedded vision hardware.	
Custom Quadruped Robot Design & Development - Rudy	Winter 2026
- Designed a 12-DOF quadruped with hip abduction for active balance, dynamic stability, and center-of-mass control. - Modeled and iterated the full mechanical system in CAD to validate joint range of motion and assembly fit. - Integrated Dynamixel actuators with TTL Bus, Ubuntu OS Raspberry Pi, and a proximity sensor into a unified system. - Built a ROS 2 C++ control stack with URDF modeling, joint position control, and kinematics for gait control.	
Franka Emika Panda - Autonomous Color Sorting	Fall 2025
- Led a team in developing a Python-based ROS 2 autonomous color-sorting pipeline for a Franka Emika robotic arm. - Calibrated robot workspace and object drop zones using AprilTags with position-averaging techniques. - Implemented vision-based object detection and color differentiation using an Intel RealSense camera with OpenCV.	

SKILLS

- Programming & Version Control** – C++, Python, C, Docker, MATLAB, SQL, Linux, Git, Unit Testing
- Robotics** – ROS 2, SLAM, Motion Planning, Computer Vision, Deep Learning, Sensor Fusion, State Estimation
- Embedded Systems**: Microcontrollers, PWM, ADC, I2C, UART, TTL, PID Control, Motor Control
- Simulation/Modeling**: CoppeliaSim, Gazebo, RViz, MATLAB-Simulink, PyBullet
- ML & Vision Libraries**: OpenCV, NumPy, PyTorch, TensorFlow
- Manufacturing & Design**: Fusion 360, SolidWorks, GD&T, 3D Printing, CNC Machining, Soldering

Leadership: Butler University Engineering Club (President, 2022), Sigma Nu (Housing Manager, 2021)